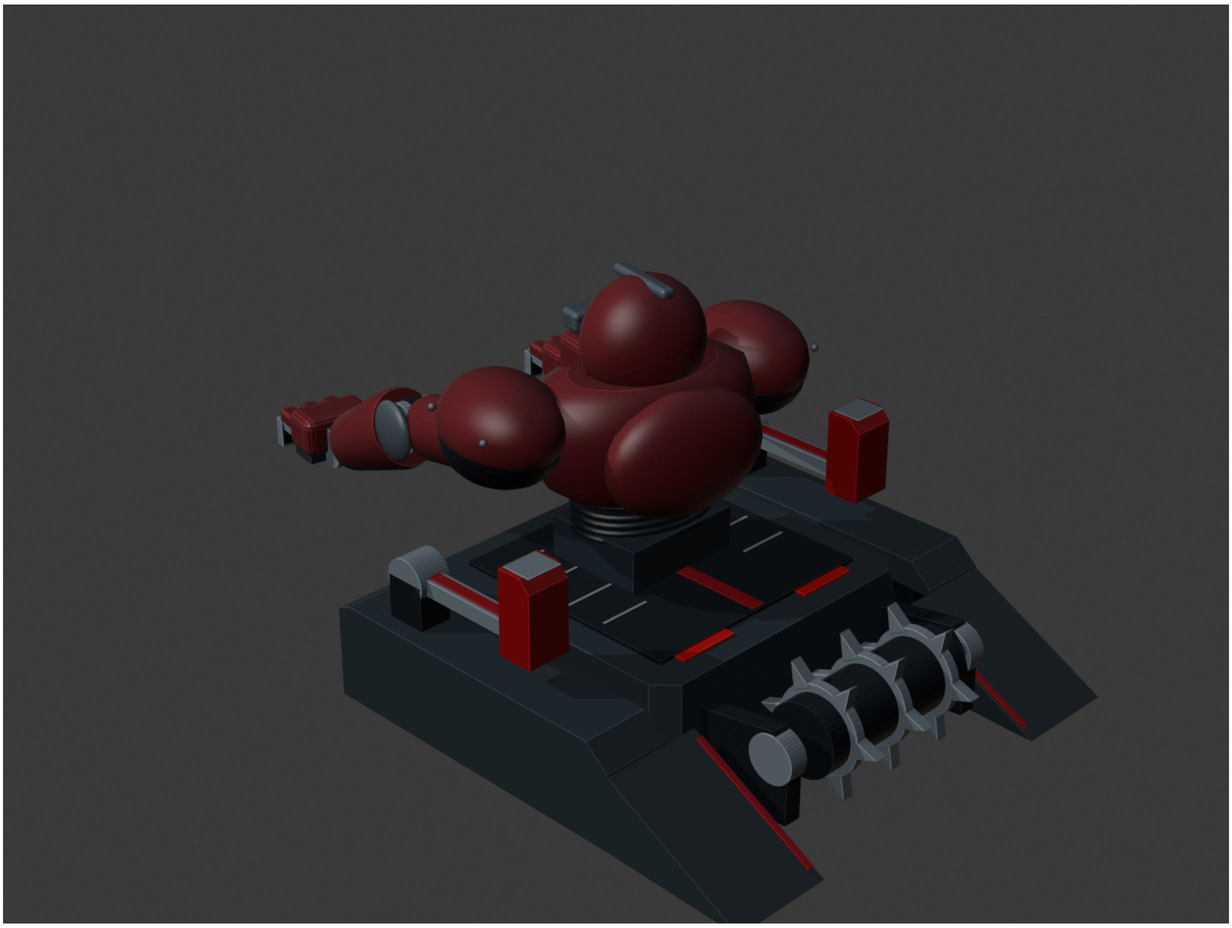
# SENTINELT

Robo Wars RC Combat Robot

## Designed and developed by Thato Glen Assegaai

Independent mechanical, control, safety and documentation development



**Verified CAD envelope: 318.16 mm x 495.00 mm x 255.41 mm**

Footprint audit: 180 / 180 frames PASS

# Robo Wars Design Requirements

Engineering limits built directly into SentinelT

- Maximum footprint: 500 mm x 500 mm.
- Robot mass: strictly below 5 kg.
- Human-operated remote control limited to 2.4 GHz.
- Accessible main ON/OFF power isolation.
- Emergency Stop mechanism that disables actuator power.
- No projectiles, flames or liquids.

**VERIFIED SENTINELT FOOTPRINT**

**318.16 mm x 495.00 mm**

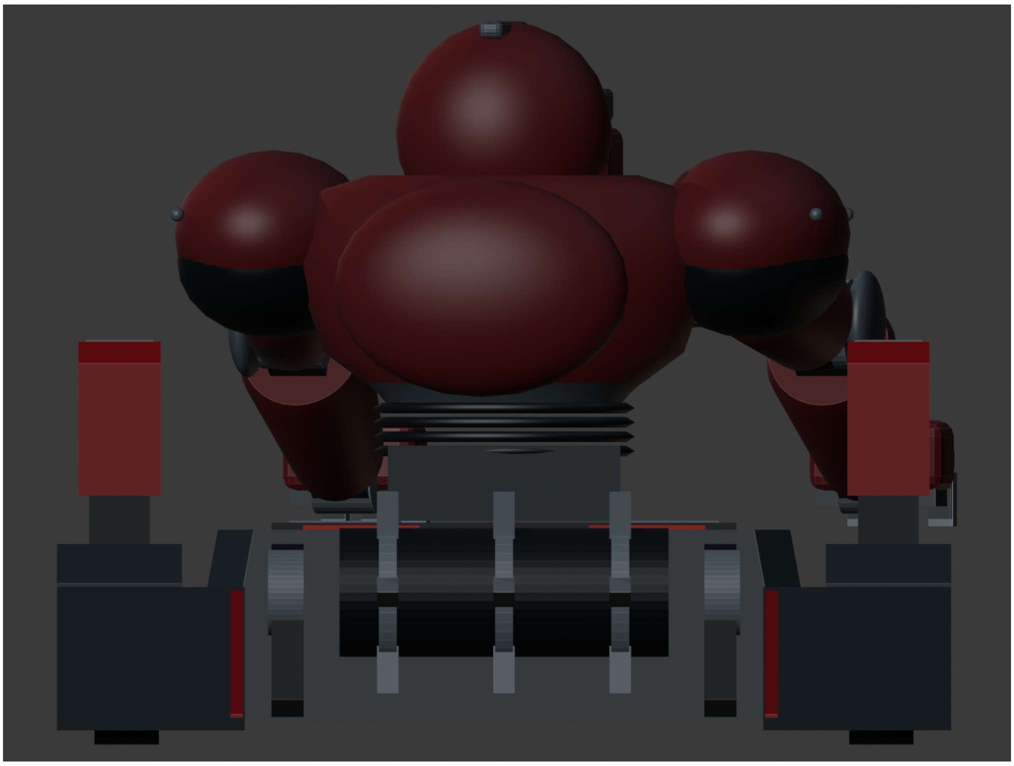
**ENGINEERING MASS ALLOCATION**

**4.650 kg**

The mass value is an engineering budget and will be physically verified after fabrication.

# SentinelT Solution & Buildability

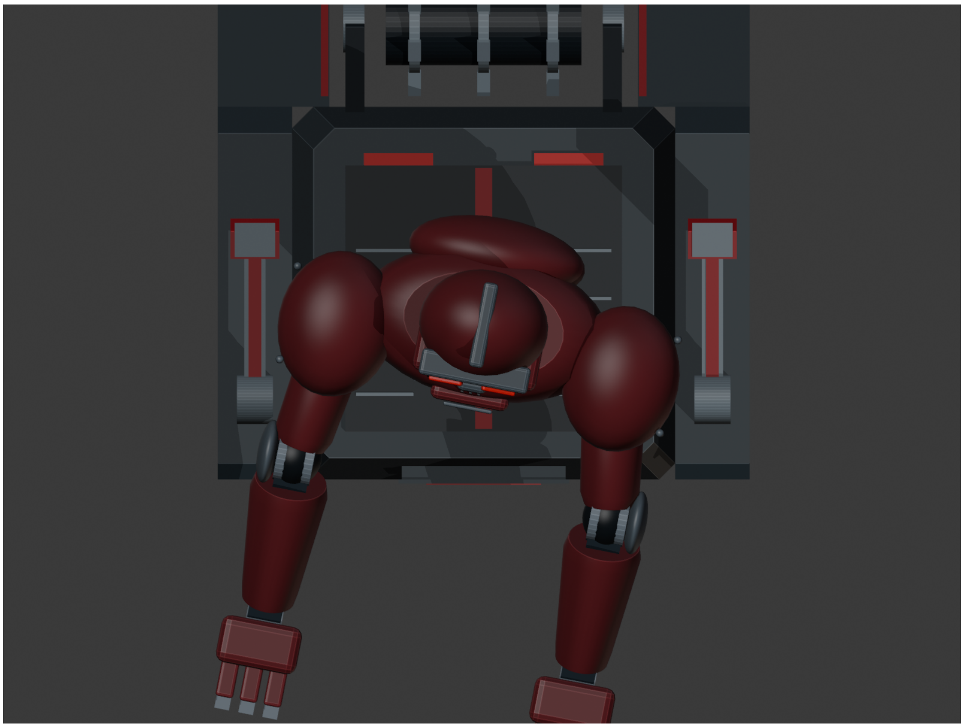
Compact, serviceable and safety-focused Robo Wars architecture



- Armored HDPE chassis with protected wheel and drive areas.
- Sloped front and side protection with removable service panels.
- Direct-drive 12 V geared motors with 125 mm wheels.
- Accessible internal packaging for battery, receiver, ESC and safety hardware.
- 2.4 GHz human-operated control with safe-state logic.
- Guarded active-mechanism provision integrated into the chassis layout.

# Mechanical CAD & Verification

Blender CAD and Python-based dimensional auditing



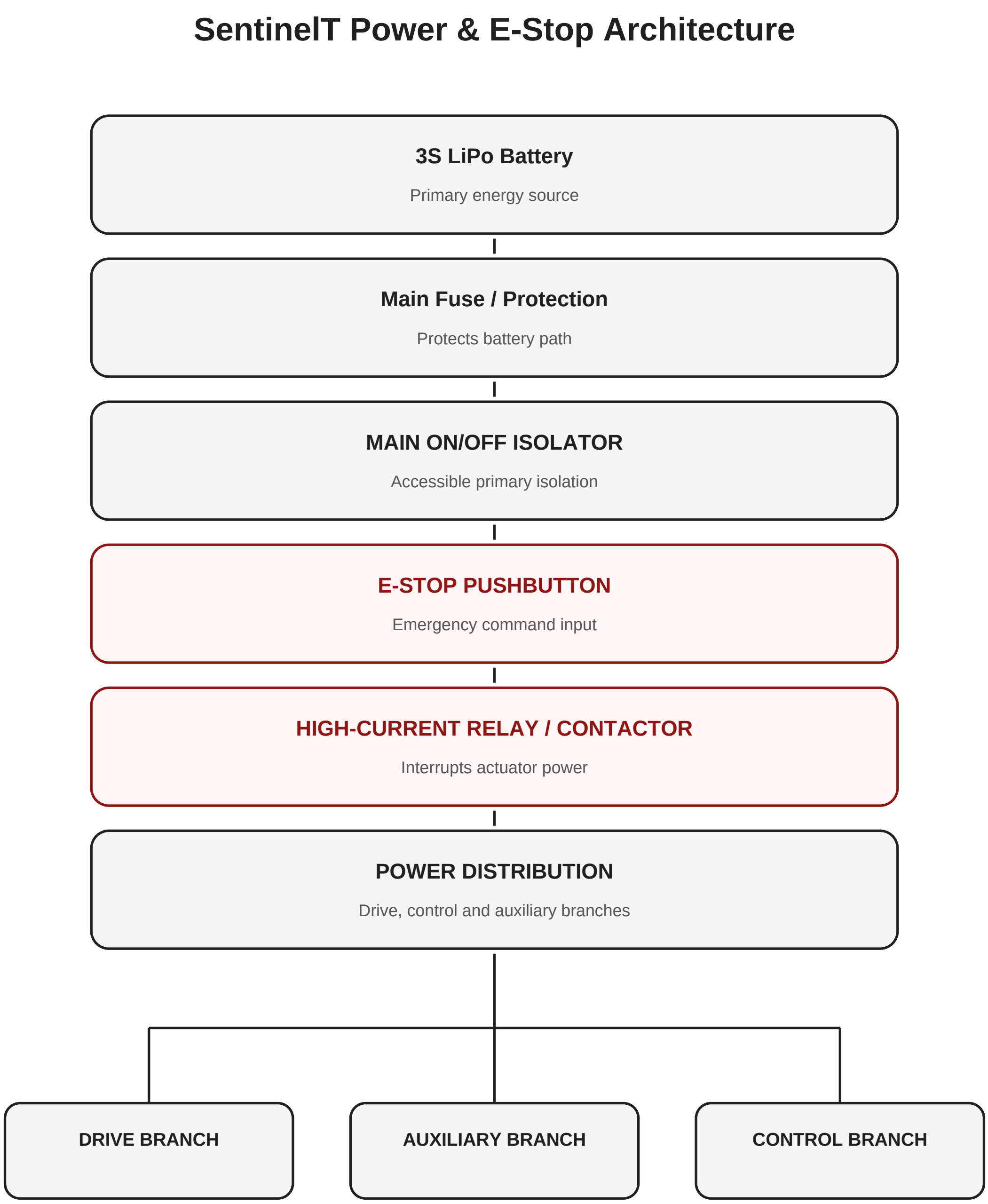
MAXIMUM CAD WIDTH (X)
**318.16 mm**

MAXIMUM CAD LENGTH (Y)
**495.00 mm**

ANIMATION-RANGE FOOTPRINT RESULT
**180 / 180 frames PASS**

# Electronics, Control & Safety

Main isolation, E-Stop and 2.4 GHz RC architecture




- 3S 11.1 V LiPo power architecture.
- 2.4 GHz receiver and safe-state control logic.
- Main ON/OFF isolation is upstream of actuator distribution.
- Emergency pushbutton commands a separate high-current isolation device.

# Mass, Components & Cost

Current engineering procurement plan

ENGINEERING MASS BUDGET

**4.650 kg**

DESIGN MARGIN BELOW 5 KG

**0.350 kg**

CURRENT PRICED COMPONENT SUBTOTAL

**R 8,039.15**

- Priced: HDPE chassis / armor stock, drive motors, wheels, RC system, battery, ESC, isolation, E-Stop hardware, power distribution, wiring and fasteners.
- The guarded active-mechanism motor / controller remains marked TBD until its mass and electrical compatibility are finalized.
- The Bill of Materials records quantity, unit cost, supplier, stock code, direct URL and total line cost.

R8,039.15 is a priced subtotal, not the final implementation total.

# Project Owner & Responsibilities

Thato Glen Assegaai

- Robo Wars system concept and engineering architecture.
- Blender CAD modelling and mechanical packaging.
- Python-based footprint and mass-budget tools.
- 2.4 GHz RC control framework and safe-state logic.
- Main ON/OFF and E-Stop safety architecture.
- Component research, Bill of Materials and procurement planning.
- Technical documentation and GitHub repository organisation.

Third-party commercial components are identified by manufacturer and supplier in the Bill of Materials.